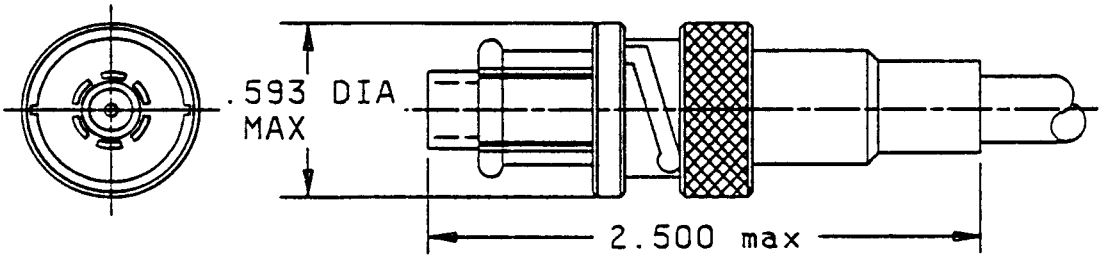
NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time the document is revised.

PERFORMANCE SPECIFICATION SHEET

CONNECTORS, COAXIAL, RADIO FREQUENCY, HIGH VOLTAGE
(SERIES SHV (CABLED) - PLUG, SOCKET CONTACT, CLASS 2)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-PRF-39012.



INCHES	MM
.593	15.06
2.500	63.50

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Dimension 2.500 defines the maximum length of the connector when assembled to the appropriate cable.
4. Dimension .593 is the largest overall diameter of the connector.
5. When required, wrench flats are to accommodate standard wrench openings per H-28, app. 10.
6. All undimensioned pictorial representations are for reference purposes only.

FIGURE 1. General configuration.

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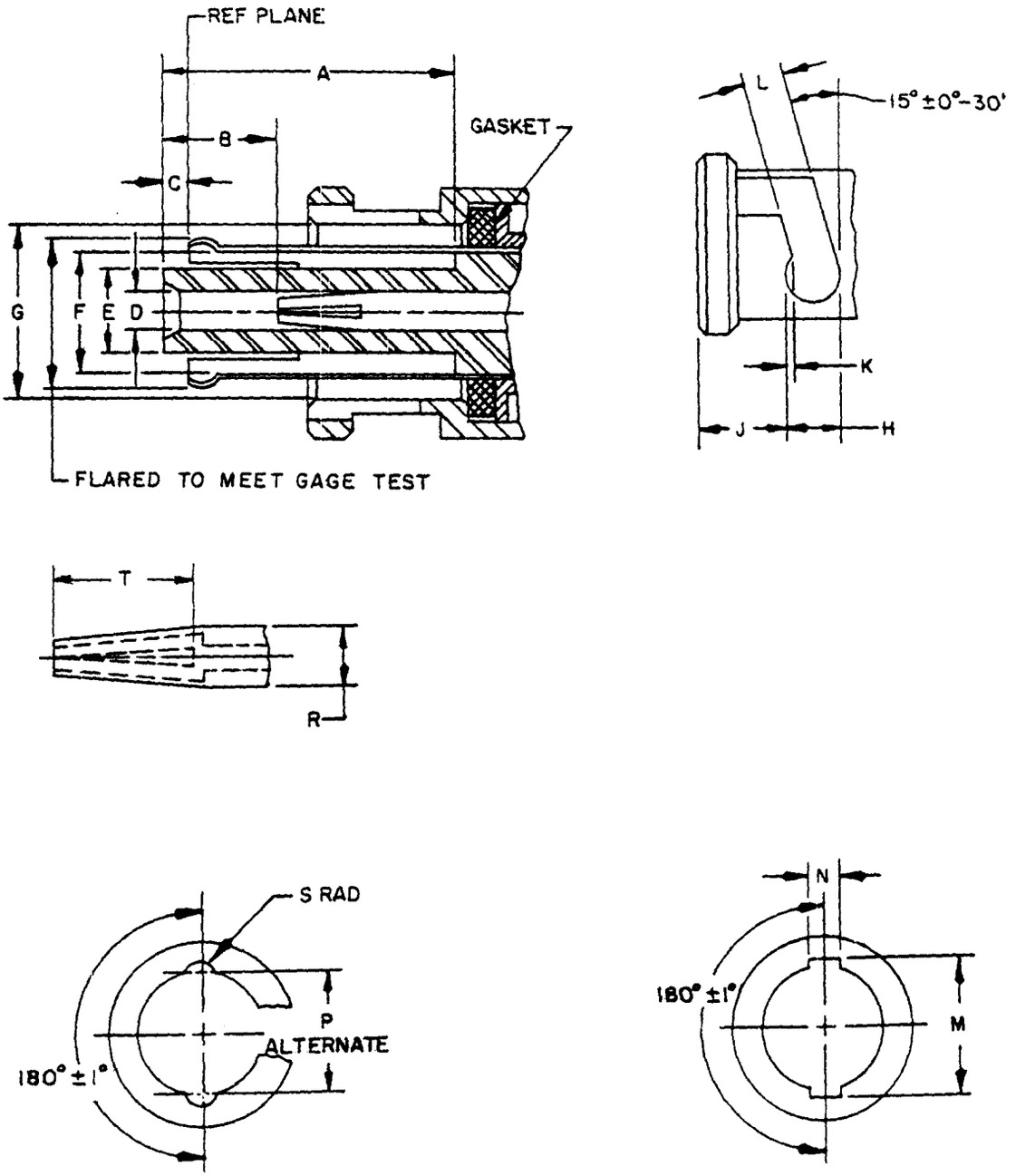


FIGURE 2. Mating dimensions for socket terminations.

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Ref	Inches		Millimeters	
	Min	Max	Min	Max
A	.628	.632	15.95	16.05
B	.238	.262	6.05	6.65
C	.046	.064	1.17	1.63
D	.082		2.08	
E	.180	.186	4.57	4.72
F	.264		6.71	
G	.385	.390	9.78	9.91
H	.124		3.15	
J	.180	.184	4.57	4.67
K	.018	.022	.46	.56
L	.091	.097	2.31	2.46
M	.463	.473	11.76	12.01
N	.091	.097	2.31	2.46
P	.394	.400	10.01	10.16
R	.081	.083	2.06	2.11
S	.045	.049	1.14	1.24
T	.214		5.44	

NOTES:
1. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.

FIGURE 2. Mating dimensions for socket terminations - Continued.

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ENGINEERING DATA:

Nominal impedance: Nonconstant.

Frequency range: Not applicable.

Voltage rating:

3,500 volts rms and 5,000 Vdc, maximum working voltage at sea level.
350 volts rms, maximum at 70,000 feet.

Temperature rating: -65° to +165°C.

REQUIRMENTS:

Design and construction: See figure 1.

Force to engage and disengage:

Longitudinal force - 3 pounds, maximum.
Torque - 2 1/2 inch-pounds, maximum.

Coupling proof torque: Not applicable.

Inspection conditions: Coupling torque - Not applicable.

Mating characteristics: See figure 2 for dimensions.

Center contact (socket):

Oversize test pin - .057 diameter minimum (nonclosed entry contacts only).
Insertion depth - .125 minimum.
Number of insertions - 1.

Insertion force test - Steel test pin diameter .054 minimum.

Test pin finish - 16 microinches.
Insertion force - 2 pounds, maximum.

Withdrawal force test: Steel test pin diameter .052 maximum.

Withdrawal force - 2 ounces, minimum.
Test pin finish - 16 microinches.

Outer contact:

Test ring ID - .319 maximum, 16 microinch finish.
Insertion force - 5 pounds maximum when inserted a minimum of .093.

Contacts with slotted members: Shall contact a .324 minimum diameter ring within .031 of their tip ends.

Center contact retention:

6 pounds minimum axial force.

Hermetic seal: Not applicable.

Leakage (pressurized connectors): Not applicable.

Insulation resistance: Method 302, test condition B, MIL-STD-202. 1 x 10^{12} ohms, minimum.

Corrosion (salt spray): Method 101, test condition B, MIL-STD-202.

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Connector durability:

500 cycles minimum at 12 cycles/minute, maximum.
The connector shall meet the mating characteristics and force to engage and disengage requirements.

Contact resistance: In milliohms maximum.

	Initial	After environment
Center contact	2.1	2.5
Outer contact	1.5	Not applicable
Braid to body	.1	Not applicable

Dielectric withstanding voltage: Method 301 of MIL-STD-202. 5,000 volts rms and 10,000 Vdc, minimum at sea level.

Vibration, high frequency: Method 204, test condition A, MIL-STD-202. No discontinuity permitted.

Shock: Method 213 of MIL-STD-202, test condition I. No discontinuity permitted.

Thermal shock : Method 107, test condition B, MIL-STD-202, except test high temperature shall be +85°C. High temperature shall be +200°C for connectors using +200°C cable.

Humidity: Method 103, test condition B per MIL-STD-202. No measurements at high humidity. Insulation resistance shall be at least 10^{12} ohms within 5 minutes after removal from humidity.

Corona level:

Voltage - 500 volts, minimum.
Altitude - 70,000 feet.

RF high potential withstanding voltage:

Voltage and frequency - Not applicable.
Leakage current - Not applicable.

Cable retention force:

Noncrimp assemblies - 40 pounds, minimum.
Crimp assemblies:

50 lbs minimum for cables .155 - .189 OD.
60 lbs minimum for cables .190 - .229 OD.
75 lbs minimum for cables .230 - .249 OD.
90 lbs minimum for cables .250 - and larger.

Coupling mechanism retention force: 100 pounds, minimum.

Part number: M39012/105- (dash number from table I).

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TABLE I. Dash number and applicable cable.

Dash no.	Applicable cable no. 3/
CATEGORY A - Field serviceable (no special tool required)	
1/ 0001 (Contacts with gold finish)	M17/29 - RG59 * M17/30 - RG62 M17/90 - RG71 M17/97 - RG210
2/ 0002 (Contacts with silver finish)	M17/29 - RG59 * M17/30 - RG62 M17/90 - RG71 M17/97 - RG210

NOTES:
1/ Cable to be used when performing tests requiring cable.
2/ Gold plated contacts: Minimum thickness of 0.00005 inch in accordance with MIL-G-45205, type II, class I with suitable underplate.
2/ Silver plated contacts: Minimum thickness of 0.0002 inch.
3/ See MIL-C-17 for working voltage of cables.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Navy - MI
Army -
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AR, AT, ME
Navy - AS, MC, OS
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

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